

UNIFORMIZATION AND FUCHSIAN MODELS OF HYPERBOLIC RIEMANN SURFACES

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ABSTRACT. We use the Uniformization Theorem and universal coverings of Riemann surfaces to classify Riemann surfaces as a quotient. Then, we define Fuchsian groups and prove that every hyperbolic Riemann surface is the quotient of the upper half plane by a Fuchsian group, which gives rise to Fuchsian models of hyperbolic Riemann surfaces.

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1. INTRODUCTION

Riemann surfaces appear in various contexts in math such as algebraic geometry (as we saw when we proved Riemann-Roch) and number theory, as well as physics, in areas such as string theory or conformal field theory. It thus compels us to look for ways to classify Riemann surfaces and understand their structure.

A particularly nice example of a Riemann surface is the torus, which can be obtained as a quotient $\mathbb{C}/\Lambda$ where Λ is a rank 2 lattice in $\mathbb{C}$. The torus is especially nice since it is compact, but in general it let's us work with just a quadrilateral in the complex plane, and we would like to ask if we can do something similar for other Riemann surfaces.

We use Uniformization Theorem to classify simply connected Riemann surfaces. Then, using some algebraic topology, we show every Riemann surface has a simply connected Riemann surface as a cover (i.e. a universal cover), which will allow us to classify Riemann surfaces as elliptic, parabolic, or hyperbolic.

The hyperbolic case turns out to be the most interesting, and we show that every hyperbolic Riemann surface is actually the quotient of the upper half plane by what is called a Fuchsian group.

2. UNIFORMIZATION AND UNIVERSAL COVERS OF RIEMANN SURFACES

2.1. The Uniformization Theorem. Recall that a Riemann surface is a complex 1-manifold, which for simplicity we assume is connected:

Definition 2.1. A **Riemann surface** is a connected Hausdorff topological space X equipped with a holomorphic atlas $(U_\alpha, \varphi_\alpha)_{\alpha \in I}$ satisfying

- (i) $\{U_\alpha\}_{\alpha \in I}$ is an open covering of X .
- (ii) $\varphi_\alpha : U_\alpha \rightarrow \mathbb{C}$ is a homeomorphism from U_α to an open subset of $\mathbb{C}$.
- (iii) The transition functions $\varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \rightarrow \mathbb{C}$ are holomorphic.

Remark 2.2. Usually we require manifolds to be second-countable but a theorem by Radó says every connected Riemann surface is second-countable. (See Forster section 23.)

We can then define all the usual notions of holomorphicity and conformality through taking charts:

Definition 2.3. Let X, Y be Riemann surfaces. A function $f : X \rightarrow Y$ is said to be **holomorphic** if it is holomorphic in charts, i.e. if X has an atlas $(U_\alpha, \varphi_\alpha)_{\alpha \in I}$ and Y has an atlas $(V_\beta, \phi_\beta)_{\beta \in J}$, then $\phi_\beta \circ f \circ \varphi_\alpha^{-1}$ is holomorphic as a complex function. Similarly we say f is **conformal** if it is holomorphic with a holomorphic inverse.

It turns out we have a nice classification of simply connected Riemann surfaces, which in this paper we do not prove:

Theorem 2.4 (Uniformization Theorem). *Let X be a simply connected Riemann surface. Then X is conformally equivalent (there exists a conformal map) to either the upper half plane $\mathbb{H}$, the complex plane $\mathbb{C}$, or the Riemann sphere $\hat{\mathbb{C}}$.*

This can be thought of as the Riemann surface equivalent of the Riemann mapping theorem, which says that any simply connected domain in $\mathbb{C}$ which is not the entire complex plane is conformally equivalent to the unit disk. (Many books state the uniformization theorem with the unit disk instead of the upper half plane, but they are conformally equivalent by the Riemann mapping theorem.)

However, to classify Riemann surfaces which are not simply connected, we will need tools from algebraic topology.

2.2. Some Algebraic Topology. We develop some of the basic algebraic topology needed to discuss universal covers (mainly since I have not taken a course in the subject, like Math 225C).

Let X be a topological space.

Definition 2.5 (Homotopy of paths). Given paths $\gamma_0, \gamma_1 : [0, 1] \rightarrow X$ with the same endpoints, a **homotopy** between γ_0 and γ_1 is a continuous function $F : [0, 1] \times [0, 1] \rightarrow X$ with $F(\cdot, 0) = \gamma_0$ and $F(\cdot, 1) = \gamma_1$ satisfying $F(0, t) = \gamma_0(0) = \gamma_1(0)$ and $F(1, t) = \gamma_0(1) = \gamma_1(1)$ for all t . In this case we say γ_0 and γ_1 are **homotopic**.

Definition 2.6 (Homotopy class). Define an equivalence relation on paths in X where $\gamma \sim \tilde{\gamma}$ if they are homotopic. The equivalence classes, denoted $[\gamma]$, are called **homotopy classes**.

Remark 2.7. Given paths α, β, γ and homotopies F between α and β and G between β and γ , define

$$H(s, t) = \begin{cases} F(s, 2t) & t \in [0, 1/2] \\ G(s, 2t - 1) & t \in [1/2, 1] \end{cases}$$

Then H is a homotopy from α to γ and transitivity follows. Reflexivity and symmetry obvious, so this is indeed an equivalence relation.

Proposition 2.8. Define a concatenation of paths $\alpha, \beta : [0, 1] \rightarrow X$ by

$$\alpha \cdot \beta(s) = \begin{cases} f(2s) & s \in [0, 1/2] \\ g(2s - 1) & s \in [1/2, 1] \end{cases}$$

Now define a composition on the homotopy classes by $[\alpha][\beta] := [\alpha \cdot \beta]$. We also define the inverse path $\bar{\alpha}(s) = \alpha(1 - s)$. Then let the set of homotopy classes of closed paths in X with base point x_0 be denoted by $\pi_1(X, x_0)$. This is a group with respect to the composition.

The proof is omitted in the interest of time. It mainly involves reparameterizing a bunch of paths.

Proposition 2.9. Given two base points x_0 and x_1 and a path γ from x_0 to x_1 , define $\varphi : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ by $\varphi([\alpha]) = [\bar{\gamma} \cdot \alpha \cdot \gamma]$. Then φ is an isomorphism.

Proof. Given $[\alpha], [\beta] \in \pi_1(X, x_0)$, we have $\varphi([\alpha \cdot \beta]) = [\bar{\gamma} \cdot \alpha \cdot \beta \cdot \gamma] = [\bar{\gamma} \cdot \alpha \cdot \gamma][\bar{\gamma} \cdot \beta \cdot \gamma] = \varphi([\alpha])\varphi([\beta])$. Define the inverse by $\varphi^{-1}([\alpha]) = [\gamma \cdot \alpha \cdot \bar{\gamma}]$ and it is easy to check this is homomorphism as well, and is the inverse. $\square$

Definition 2.10. $\pi_1(X, x_0)$ is called the **fundamental group** of X with base point x_0 . Because of the invariance under base point, we can also just write $\pi_1(X)$.

Definition 2.11. Let $X, \tilde{X}$ be topological spaces. We say a continuous map $p : \tilde{X} \rightarrow X$ is a **covering map** of X if each point $x \in X$ has a neighborhood U , called an **evenly covered neighborhood**, such that $p^{-1}(U)$ is the union of disjoint open sets (called **sheets**) $\{V_\alpha\}_{\alpha \in I}$ such that $p|_{V_\alpha}$ is a homeomorphism onto U . We call $\tilde{X}$ the **covering space** and X the **base space**. The collection of sets $p^{-1}(x)$ are called the **fibers** of p .

Definition 2.12. Let X, Y be topological spaces. Given a covering map $p : \tilde{X} \rightarrow X$ and a continuous map $f : Y \rightarrow X$, we say $\tilde{f} : Y \rightarrow \tilde{X}$ is a **lift** of f under p if the following diagram commutes:

$$\begin{array}{ccc} & & \tilde{X} \\ & \nearrow \tilde{f} & \downarrow p \\ Y & \xrightarrow{f} & X \end{array}$$

Theorem 2.13 (Homotopy Lifting). Let $p : \tilde{X} \rightarrow X$ be a covering map and let $F : [0, 1] \times [0, 1] \rightarrow X$ be a homotopy of paths. Given a lift $\tilde{\gamma}_0 : [0, 1] \rightarrow \tilde{X}$ of $\gamma_0 : [0, 1] \rightarrow X$, where $\gamma_0 = F(\cdot, 0)$, there exists a unique homotopy $\tilde{F} : [0, 1] \times [0, 1] \rightarrow \tilde{X}$ which lifts F .

Remark 2.14. We omit the proof of this theorem since it was mostly proved in Math 246B. We looked at the special case of the exp map, but only used its covering map properties, and the rest of the proof can be almost used as is.

The same proof also gives tells us we can lift paths:

Theorem 2.15 (Path Lifting). Let $p : \tilde{X} \rightarrow X$ be a covering map and γ a path in X . Given $\tilde{x} \in \tilde{X}$ there exists a unique lift of γ to a path $\tilde{\gamma}$ in $\tilde{X}$ such that $\tilde{\gamma}(0) = \tilde{x}$.

We would also like to have a nice condition for which we know a map is a covering map, which will be useful later when we explicitly construct a covering map of Riemann surfaces. To do this we need a more general theorem about lifts:

Theorem 2.16. *Suppose X, Y are Hausdorff spaces and $p : Y \rightarrow X$ is a local homeomorphism with the path lifting property. Suppose Z is a locally simply connected space and $f : Z \rightarrow X$ is a continuous map. Then given $z_0 \in Z$ and $y_0 \in Y$ such that $f(z_0) = p(y_0)$, there exists a unique lift of f to $\tilde{f} : Z \rightarrow Y$ such that $\tilde{f}(z_0) = y_0$.*

Remark 2.17. This theorem is usually stated with p as a covering map, but as explained above we wish to use this to show when a map is a covering map, so we use the bare minimum necessary. We again omit the proof in the interest of time, but it is proved in a similar spirit to Lemma 2.23 (see below), where $\tilde{f}$ is constructed through a lift of paths, and the continuity of f is shown using the fact p is a local homeomorphism and the uniqueness of path lifts.

Theorem 2.18. *Suppose X a locally simply connected Hausdorff space and Y is a Hausdorff space, and $p : Y \rightarrow X$ is a local homeomorphism with the path lifting property. Then p is a covering map.*

Proof. Fix some $x \in X$ and let $p^{-1}(x) = \{y_j\}_{j \in J}$ be its fiber. Let U be a simply connected neighborhood of x , which itself is locally simply connected, and let $f : U \rightarrow X$ be the inclusion map. Then by the previous theorem there exists unique lifts $\tilde{f}_j : U \rightarrow Y$ such that $\tilde{f}_j(x) = y_j$. Let $V_j = \tilde{f}_j(U)$. We claim

$$p^{-1}(U) = \bigcup_{j \in J} V_j$$

For the $\supseteq$ direction we note that since $\tilde{f}_j$ is a lift we have $p \circ \tilde{f}_j = f$ and since f is just the inclusion map we have $p(V_j) = p(\tilde{f}_j(U)) = f(U) = U$. Thus $V_j \subseteq p^{-1}(U)$.

Now for the $\subseteq$ direction, take $y \in p^{-1}(U)$ and let $u = p(y)$. Consider a path γ in U from x to u . We lift the inverse of this path so that it starts at y . Inverting the path again, we can check this is a lift of γ starting at some preimage y_j . But note that $\tilde{f}_j \circ \gamma$ is also a lift, and by uniqueness they are equal. Hence the endpoint of the lift $y \in V_j$.

The fact that the V_j 's are disjoint follows from the fact that the $\tilde{f}_j$'s are lifts and are unique up to fixing some point. Note we have by definition $p \circ \tilde{f}_j = f = \text{id}_U$. We also have $\tilde{f}_j \circ p|_{V_j} = \text{id}_{V_j}$ since given $y \in V_j$ we can always find $x \in U$ such that $\tilde{f}_j(x) = y$. Then $\tilde{f}_j(p|_{V_j}(y)) = \tilde{f}_j(p|_{V_j} \circ \tilde{f}_j(x)) = \tilde{f}_j(x) = y$. Hence $p|_{V_j} : V_j \rightarrow U$ is a homeomorphism, and p is a covering map. $\square$

Definition 2.19. If X is path connected and locally path connected, then a covering map $p : \tilde{X} \rightarrow X$ is called the **universal cover** if $\tilde{X}$ is simply connected.

Remark 2.20. Under the above assumptions on X it is possible to show that map between fundamental groups induced by covering maps satisfies a Galois-type correspondence (like with field extensions and Galois groups), and in this sense the universal cover is named so since it is a covering of any other cover. A universal cover is also unique up to isomorphism, hence why we call it *the* universal cover.

One way to think about universal covers in the context of Riemann surfaces is that it gives a nice way to build a simply connected surface that maps into the base space. For example, the Riemann surface of the log function is often visualized as a spiral, glued together with infinitely many copies of the complex plane, which is the universal cover of the punctured plane.

Definition 2.21. Given a covering map $p : \tilde{X} \rightarrow X$ a **deck transformation** is a homeomorphism $f : \tilde{X} \rightarrow \tilde{X}$ such that the following diagram commutes:

$$\begin{array}{ccc} \tilde{X} & \xrightarrow{f} & \tilde{X} \\ & \searrow p & \downarrow p \\ & & X \end{array}$$

The group of deck transformations of $\tilde{X}$ is denoted $G(\tilde{X})$.

Note that given a fiber $p^{-1}(x)$ the deck transformation must satisfy $p(f(p^{-1}(x))) = p(p^{-1}(x)) = \{x\}$. Hence f must map $p^{-1}(x)$ into $p^{-1}(x)$. In this sense we can call deck transformations “fiber-preserving maps.” Under some more assumptions about the covering (which we omit discussion about) the fibers must also map bijectively into each other, and it happens to be true for universal covers. Then the deck transformations just permute the points in the fibers.

Now recall the covering space contains many sheets of evenly covered neighborhoods in the base space, and the all the sheets of a given evenly covered neighborhood can be thought of as just the union of the fibers of the neighborhood. (If you think of stacking the sheets on top of each other like paper, then slicing them vertically gives you all the fibers.) Then the deck transformations can be thought of as permuting the sheets, so it is natural to think that by quotienting out by the deck transformations, we recover the base space:

Theorem 2.22. *Suppose X is path connected, locally path connected, and has a universal cover $p : \tilde{X} \rightarrow X$. Then $\tilde{X}/G(\tilde{X})$ is homeomorphic to X .*

We will first prove a useful lemma that says the group of deck transformations acts transitively on the fibers:

Lemma 2.23. *Suppose X is path connected, locally path connected, and has a universal cover $p : \tilde{X} \rightarrow X$. Given a fiber $p^{-1}(x)$ and $\tilde{x}_1, \tilde{x}_2 \in p^{-1}(x)$ there exists a deck transformation $g \in G(\tilde{X})$ such that $g(\tilde{x}_1) = \tilde{x}_2$.*

Proof. For any $\tilde{y} \in \tilde{X}$, take a path α from $\tilde{x}_1$ to $\tilde{y}$ (since $\tilde{X}$ is simply connected). Then define $\gamma = p \circ \alpha$ so $\gamma(0) = x$. Then lift it to a curve $\tilde{\gamma}$ such that $\tilde{\gamma}(0) = \tilde{x}_2$. Define $g(\tilde{y}) = \tilde{\gamma}(1)$.

We need to show g is well-defined, i.e. it does not depend on the choice of path from $\tilde{x}_1$ to $\tilde{y}$. Let α, β be two such paths. Then $\alpha \cdot \beta^{-1}$ is null-homotopic by simple connectivity, and hence α and β are homotopic (we omit this proof). Since p is a covering map, it is a local homeomorphism and hence $p \circ \alpha$ and $p \circ \beta$ are also homotopic with fixed starting and end points. Now lift these two paths back to paths in $\tilde{X}$ starting at $\tilde{x}_2$. By the homotopy lifting theorem, they must still have the same endpoints, so g is well-defined.

By taking constant paths, we clearly see $g(\tilde{x}_1) = \tilde{x}_2$. By definition we also have $p(g(\tilde{y})) = (p \circ \alpha)(1) = p(\tilde{y})$.

Now we show p is a local homeomorphism. Since p is a covering map and X is locally path connected, consider a *path-connected* evenly covered neighborhood U in X , and a sheet V (which must also be path-connected) so $p|_V : V \rightarrow U$ is a homeomorphism. Then fix $\tilde{y}_0 \in V$, and let $\tilde{z}_0 := g(\tilde{y}_0)$. Since $p \circ g = p$ we must have $z_0 \in p^{-1}(U)$ as well, so $z_0 \in V'$ for some sheet V' . Then $p|_{V'} : V' \rightarrow U$ is also a homeomorphism. We claim that g maps V into V' , and that

$$g|_V = (p|_{V'})^{-1} \circ p|_V$$

Fix $\tilde{y} \in V$. Consider a path α in $\tilde{X}$ from $\tilde{x}_1$ to $\tilde{y}_0$ and a path γ in V from $\tilde{y}_0$ to $\tilde{y}$. Then by the path lifting property we can lift $p \circ (\alpha \cdot \gamma)$ to a path in $\tilde{X}$ starting at $\tilde{x}_2$. In particular this lift is the concatenation of the lift of $p \circ \alpha$ starting at $\tilde{x}_2$ and ending at $g(\tilde{y}_0) = \tilde{z}_0$ and the lift of $p \circ \gamma$ starting at $\tilde{z}_0$ and ending at $g(\tilde{y})$. But note $(p|_{V'})^{-1}(p \circ \gamma)$ is the unique lift by the path lifting property, so it must end at $(p|_{V'})^{-1}(p(\tilde{y}))$ so we have proved the equation above. Thus g is a local homeomorphism.

We can clearly construct an inverse of g by just swapping $\tilde{x}_1$ and $\tilde{x}_2$. We check that $h \circ g = \text{id}$ and the other side follows similarly. Given $\tilde{y} \in \tilde{X}$, pick a path α from $\tilde{x}_1$ to $\tilde{y}$, so $g(\tilde{y})$ is the endpoint of the lift of $p \circ \alpha$, call it $\tilde{\alpha}$. But then $h(g(\tilde{y}))$ is the end point of the lift of $p \circ \tilde{\alpha} = p \circ \alpha$ starting at $\tilde{x}_1$, but by uniqueness that lift is α , which ends at $\tilde{y}$. Thus $h(g(\tilde{y})) = \tilde{y}$. Thus g is also a bijection, and hence must be a global homeomorphism. Thus g is the desired deck transformation. $\square$

Remark 2.24. This lemma is much cleaner to prove if we have the lifting criterion, but for the sake of remaining as self-contained as possible, we forego that route.

Proof of Theorem 2.22. By definition given $g \in G(\tilde{X})$ we have $p \circ g = p$, so given a fiber $p^{-1}(x)$ and $\tilde{x} \in p^{-1}(x)$, we must have $p(G(\tilde{X})\tilde{x}) = \{p(\tilde{x})\}$, where $G(\tilde{X})\tilde{x}$ denotes the orbit of $\tilde{x}$. Hence p is constant on $G(\tilde{X})$ -orbits, and naturally induces a continuous map on the quotient $\tilde{p} : \tilde{X}/G(\tilde{X}) \rightarrow X$.

Now $\tilde{p}$ must be bijective. Since p is surjective, so is $\tilde{p}$. Let $\pi : \tilde{X} \rightarrow \tilde{X}/G(\tilde{X})$ be the quotient map. Given $\pi(\tilde{x}_1), \pi(\tilde{x}_2) \in \tilde{X}/G(\tilde{X})$, we have $\tilde{p}(\pi(\tilde{x}_1)) = \tilde{p}(\pi(\tilde{x}_2)) \implies p(\tilde{x}_1) = p(\tilde{x}_2) \implies \tilde{x}_1, \tilde{x}_2$ lie in the same fiber, but by the lemma the fibers are exactly the orbits, so they must lie in the same orbit. Hence $\pi(\tilde{x}_1) = \pi(\tilde{x}_2)$ and $\tilde{p}$ is injective, and thus bijective.

Finally, we show that $\tilde{p}$ is a local homeomorphism. Let U be any openly covered neighborhood of X , and let V be a sheet of its preimage, so $p|_V : V \rightarrow U$ is a homeomorphism. Then for $\tilde{x}, \tilde{y} \in V$, if $\pi(\tilde{x}) = \pi(\tilde{y})$, the lie in the same orbit, and hence the same fiber, so $p(\tilde{x}) = p(\tilde{y})$. But since $p|_V$ is injective, we have $\tilde{x} = \tilde{y}$, so $\pi|_V$ is injective and automatically bijective. Now note that π is an open map since for any $W \subseteq \tilde{X}$ open we have $\pi^{-1}(\pi(W)) = \bigcup_{g \in G(\tilde{X})} g(W)$ which is open. Thus $\pi|_V$ is a homeomorphism. Since $p = \tilde{p} \circ \pi$, we then have $\tilde{p}|_{\pi(V)} = p|_V \circ (\pi|_V)^{-1}$ on V , so $\tilde{p}$ is a local homeomorphism.

Since $\tilde{p}$ is a local homeomorphism and a bijection, it is a global homeomorphism and we are done. $\square$

Definition 2.25. We say a group of homeomorphisms G acts properly discontinuously on X if each point $x \in X$ has a neighborhood U such that

$$g(U) \cap U \neq \emptyset$$

for finitely many $g \in G$.

Proposition 2.26. Let $p : \tilde{X} \rightarrow X$ be a cover. The group of deck transformations $G(\tilde{X})$ acts

(i) freely

(ii) properly discontinuously

on $\tilde{X}$.

Proof.

- (i) Suppose $g \in G(\tilde{X})$ and $g(\tilde{x}) = \tilde{x}$ for some $\tilde{x} \in \tilde{X}$. Let γ be any path in $\tilde{X}$ starting at $\tilde{x}$. Then since g is a deck transformation we have $p \circ \gamma = (p \circ g) \circ \gamma = p \circ (g \circ \gamma)$. But then both γ and $g \circ \gamma$ are lifts of $p \circ \gamma$ starting at $\tilde{x}$, so by the uniqueness part of path lifting, we must have $\gamma = g \circ \gamma$. Since we can do this for any path, g must fix every point in $\tilde{X}$, and must be the identity.
- (ii) Since p is locally homeomorphic, let V be some neighborhood of $\tilde{X}$ homeomorphic to a neighborhood U of X by p . Let $g \in G(\tilde{X})$ be s.t. $g(V) \cap V \neq \emptyset$. Then $g(\tilde{x}) = \tilde{y}$ for some $\tilde{x}, \tilde{y} \in V$. Thus they lie in the same $G(\tilde{X})$ -orbit and hence the same fiber of p as shown in the proof of Theorem 2.22. But since p is homeomorphic on V , V only intersects the fiber at one point, and hence $\tilde{x} = \tilde{y}$. Since the group of deck transformations act freely, it follows $g = e$, where e is the identity map.

$\square$

Remark 2.27. We actually get a stronger result for the group of deck transformations, but as we will see when looking at Fuchsian groups, this definition of properly discontinuous action will characterize them, and requiring $g(U) \cap U = \emptyset$ for every non-identity element is too restrictive.

2.3. Universal Covers of Riemann Surfaces. We have just seen that even in the most general case of a topological space, we can write the space as a quotient of its universal cover by a group which acts properly discontinuously. It is actually possible to show that when the base space is connected, this group must then be discrete, but for simplicity we avoid proving this and instead wait until the discussion of Fuchsian groups to give a more concrete characterization.

Now we would like upgrade these results to Riemann surfaces, and hopefully upgrade the corresponding homeomorphisms to conformal maps. It turns out the most difficult part is showing that every Riemann surface has a universal cover which is also a Riemann surface, and everything after that follows pretty easily.

Recall that manifolds are very well-behaved as topological spaces:

Proposition 2.28. *Let X be a Riemann surface. Then X is path connected and locally simply connected.*

Proof. Note that any open set in $\mathbb{C}$ is locally simply connected since you can always find balls, and since homeomorphisms preserve the property, lifting this up through the charts gives that X is locally simply connected. Then using a standard argument by fixing a starting point and considering the set that is path connected to the starting point, we can show connected and locally path connected implies path connected. $\square$

Lemma 2.29. *Let X be a Riemann surface and Y be a Hausdorff space. If $p : Y \rightarrow X$ is a local homeomorphism, there is a unique complex structure on Y such that p is locally conformal.*

Proof. Let $(U_\alpha, \varphi_\alpha)_{\alpha \in I}$ be a maximal atlas for X . Since p is a local homeomorphism, for every point in Y pick a neighborhood V_α such that $p|_{V_\alpha} : V_\alpha \rightarrow U_\alpha$ is a homeomorphism where U_α is part of a chart. Then $\phi_\alpha := \varphi_\alpha \circ p|_{V_\alpha}$ is a chart on Y . Clearly these V_α 's cover Y , and given two charts (V_α, ϕ_α) and (V_β, ϕ_β) , the transition map is given by $\phi_\alpha \circ \phi_\beta^{-1} = \varphi_\alpha \circ p|_{V_\alpha} \circ (p|_{V_\beta})^{-1} \circ \varphi_\beta^{-1} = \varphi_\alpha \circ \varphi_\beta^{-1}$ which is holomorphic on the intersection $\phi_\beta(V_\alpha \cap V_\beta)$. Hence these charts form a complex atlas for Y .

Then by definition of the atlas on Y , p locally takes charts to charts, and hence is locally conformal. To see uniqueness, consider another atlas on Y , and note that the identity map between Y with the two different atlases equipped is conformal, meaning the transition functions between the two different atlases are holomorphic. Hence the atlases are compatible and define the same complex structure. $\square$

Theorem 2.30. *Let X be a Riemann surface. Then there exists a simply connected Riemann surface $\tilde{X}$ and a holomorphic covering map $p : \tilde{X} \rightarrow X$ (i.e. $\tilde{X}$ is the universal cover of X).*

Proof. By the previous lemma it suffices to show that X has a cover by some simply-connected Hausdorff space. We construct $\tilde{X}$ as follows. Fix $x_0 \in X$. Then for any $x \in X$ let $\pi(x_0, x)$ denote the homotopy classes of curves starting at x_0 and ending at x . Then define

$$\tilde{X} := \{(x, [\alpha]) : x \in X, [\alpha] \in \pi(x_0, x)\}$$

Then define $p : \tilde{X} \rightarrow X$ by $p(x, [\alpha]) = x$. We define a topology on $\tilde{X}$ as follows: Given $(x, [\alpha]) \in \tilde{X}$, let U be a simply connected neighborhood containing x . Then we define the subset $[U, \alpha] \subseteq \tilde{X}$ as all the points $(y, [\beta])$ such that $y \in U$ and $[\beta] = [u \cdot v]$ where u is a path from x_0 to x homotopic to α and v is a path in U from x to y . This is well-defined since X is simply connected, so all choices of v are homotopic to each other. Now let $\mathcal{B}$ be the collection of all such sets $[U, \alpha]$.

Claim (i): $\mathcal{B}$ is a base for a topology on $\tilde{X}$.

The set of $[U, \alpha]$ clearly cover $\tilde{X}$. Given $[U, \alpha]$ and $[V, \beta]$ where α goes from x_0 to x and β goes from x_0 to y , suppose $(z, [\gamma]) \in [U, \alpha] \cap [V, \beta]$. Then pick a simply connected neighborhood W of z such that $W \subseteq U \cap V$ by taking intersections if needed. We then wish to show $[W, \gamma] \subseteq [U, \alpha] \cap [V, \beta]$.

Given $(w, [\delta]) \in [W, \gamma]$, we have δ is the concatenation of a path from x_0 to z homotopic to γ and a path in W from z to w . But the second path is also a path in U and in V . Now since $(z, [\gamma]) \in [U, \alpha]$ we can write γ as the concatenation of a path from x_0 to x and a path in U from x to z . Then by associativity we see that $(w, [\delta]) \in [U, \alpha]$. This can be similarly shown for $[V, \beta]$, so the inclusion is proved. Thus $\mathcal{B}$ is a base for a topology.

Claim (ii): p is a local homeomorphism.

It suffices to show $p|_{[U, \alpha]}$ is a homeomorphism onto U for each $[U, \alpha] \in \mathcal{B}$. Clearly given $V \subseteq U$ open its preimage under $p|_{[U, \alpha]}$ is just $[V, \alpha]$. Now for any $V \subseteq U$ we have $[V, \alpha]$ is open in $[U, \alpha]$ and maps to V . Thus $p|_{[U, \alpha]}$ is a local homeomorphism. Clearly $p|_{[U, \alpha]}$ is surjective. Since any two paths with the same starting and endpoints are homotopic in U (simply connected), each $u \in U$ has exactly one preimage, since there is only homotopy class of paths from x_0 to u passing through x . Hence $p|_{[U, \alpha]}$ is bijective, and thus a homeomorphism.

Claim (iii): $\tilde{X}$ is Hausdorff.

Two points $(x, [\alpha])$ and $(y, [\beta])$ where $x \neq y$ clearly have disjoint neighborhoods since X is Hausdorff. Thus it remains to show $(x, [\alpha])$ and $(x, [\beta])$ where $[\alpha] \neq [\beta]$ have disjoint neighborhoods. Let α and β go from x_0 to x . Suppose for contradiction there is some simply connected neighborhood U of x such that $[U, \alpha] \cap [U, \beta] \neq \emptyset$, so some $(z, [\gamma])$ is in the intersection. But then γ is the concatenation of a path from x_0 to x which is homotopic to both α and β and another path from x to z , a contradiction.

Claim (iv): $\tilde{X}$ is path-connected and p has the path lifting property.

We do this in two steps. We first show we can lift any curve starting at x_0 in a way that the lifted curve ends at any point in $\tilde{X}$, and hence it is path connected. Then we lift any curve starting at any point.

Let $\alpha : [0, 1] \rightarrow X$ be a curve starting at x_0 . Define $\alpha_t : [0, 1] \rightarrow X$ by $\alpha_t(s) = \alpha(ts)$ for $t \in [0, 1]$. Given any closed curve β starting and ending at x_0 , we claim the mapping

$$\tilde{\alpha} : [0, 1] \rightarrow \tilde{X}, \quad \tilde{\alpha}(t) = (\alpha(t), [\beta \cdot \alpha_t])$$

is continuous and a lift of α starting at $(x_0, [\beta])$. The fact it is a lift is obvious. For continuity it suffices to show it for elements of the base. Fix any $t_0 \in [0, 1]$ and let U be a simply connected neighborhood of $\alpha(t_0)$. Consider any $\tilde{t} \in \tilde{\alpha}^{-1}([U, \alpha(t_0)])$. By the continuity of α for any t sufficiently close to $\tilde{t}$ we have $\alpha(t) \in U$. Furthermore by choosing t sufficiently close we can also guarantee the portion of α between t and $\tilde{t}$ lies in U . Then clearly $(\alpha(t), [\beta \cdot \alpha_t]) \in [U, \alpha(t_0)]$ for sufficiently close t . Thus the preimage is open, and hence the path is continuous.

Now given any $(x, [\alpha]) \in \tilde{X}$ consider the path from x_0 to x , and take lift α to a curve $\tilde{\alpha}$ which starts at (x_0, e) , where e is the homotopy class of constant curves. Then $\tilde{\alpha}(1) = (x, [\alpha])$ so $\tilde{X}$ is path-connected.

Now let γ be a curve with arbitrary starting point x , α a curve from x_0 to x . Then the lift $\alpha \cdot \gamma$ starting at (x_0, e) is the concatenation of the lift of α starting at (x_0, e) and ending at (x, α) , and lift of γ starting at (x, α) . Thus we have a lift of γ . Uniqueness of lifts follows as in the proof of the path lifting theorem.

Note that claims (ii), (iii), and (iv) along with Theorem 2.18 implies p is a covering map. In particular we can use homotopy lifting.

Claim (v): $\tilde{X}$ is simply connected.

Let γ be a closed path with starting and end point (x_0, e) . Then $\alpha := p \circ \gamma$ is a closed path in X with starting point x_0 . Then lift this curve as in claim (iv) with $[\beta] = e$ to get a curve $\tilde{\alpha}$ in $\tilde{X}$ which starts at (x_0, e) and ends at $(x_0, [\alpha])$. But since lifts are unique, this must coincide with γ , which ends at (x_0, e) . Hence $[\alpha] = e$, so α is null-homotopic. By homotopy lifting, γ must also be null-homotopic. By the invariance of the fundamental group under the base point we conclude $\tilde{X}$ is simply connected. $\square$

Remark 2.31. The intuition for the construction of $\tilde{X}$ is that you fix some arbitrary base point for the fundamental group, and then associate all the possible paths up to homotopy for each point of X , but you make sure the space is simply connected by equipping it with a topology using simply connected neighborhoods so that each homotopy class gets its own sheet over the neighborhood.

This now let's us classify all Riemann surfaces by its universal cover, using the Uniformization theorem.

Definition 2.32. Let X be a Riemann surface and $\tilde{X}$ be its universal cover. We say X is

- **elliptic** if its universal cover is conformally equivalent to the Riemann sphere $\hat{\mathbb{C}}$.
- **parabolic** if its universal cover is conformally equivalent to the complex plane $\mathbb{C}$.
- **hyperbolic** if its universal cover is conformally equivalent to the upper half plane $\mathbb{H}$.

Now we can use this to upgrade all the results related to deck transformations to conformal automorphisms.

Proposition 2.33. *Let X be a Riemann surface and $\tilde{X}$ be its universal cover, so the covering map $p : \tilde{X} \rightarrow X$ is locally conformal. Then the deck transformations are conformal. In other words $G(\tilde{X}) \subseteq \text{Aut}(\tilde{X})$, where $\text{Aut}(\tilde{X})$ denotes the group of conformal automorphisms of $\tilde{X}$.*

Proof. Given a deck transformation $g : \tilde{X} \rightarrow \tilde{X}$ we have $p \circ g = p$. Since p is locally conformal, for each $\tilde{x} \in \tilde{X}$ there exists a neighborhood U of $\tilde{x}$ and V of $g(\tilde{x})$ such that $p|_U$ and $p|_V$ are conformal onto the same open set $W \subseteq X$ (recall deck transformations permute the sheets of an evenly covered neighborhood). Then $g|_U = (p|_V)^{-1} \circ p|_U$, so g is locally conformal. Since g is homeomorphism and is thus bijective, g must be globally conformal. $\square$

Proposition 2.34. *Let X be Riemann surface and $\tilde{X}$ its universal cover. $\tilde{X}/G(\tilde{X})$ can be given a unique complex structure such that the quotient map $\pi : \tilde{X} \rightarrow \tilde{X}/G(\tilde{X})$ is locally conformal.*

Proof. We first show $\tilde{X}/G(\tilde{X})$ is Hausdorff. Take $\tilde{x}, \tilde{y} \in \tilde{X}$ such that $\pi(\tilde{x}) \neq \pi(\tilde{y})$. This means the orbits of $\tilde{x}$ and $\tilde{y}$ are disjoint, and they lie in different fibers. Let $x = p(\tilde{x})$ and $y = p(\tilde{y})$, so $x \neq y$. Then by the Hausdorffness of X choose disjoint neighborhoods of x and y , W_x and W_y , respectively, intersecting if needed so they are also evenly covered. Then let U be the sheet over W_x containing $\tilde{x}$ and V be the sheet over W_y containing $\tilde{y}$.

Note that since $G(\tilde{X})$ permutes the sheets, $g(U)$ is just another sheet over W_x for each $g \in G(\tilde{X})$. p restricted to each of these sheets is homeomorphic to W_x , and p restricted to V is homeomorphic to W_y . Then we must have $g(U) \cap V = \emptyset$ for all $g \in G(\tilde{X})$ or else by applying p we would have $W_x \cap W_y \neq \emptyset$, which would be a contradiction.

Now define $\bar{U} = \bigcup_{g \in G(\tilde{X})} g(U)$ and $\bar{V} = \bigcup_{g \in G(\tilde{X})} g(V)$. We clearly have $\bar{U} \cap \bar{V} = \emptyset$, but since $\bar{U}$ is $G(\tilde{X})$ -invariant, we can just apply any element $g \in G(\tilde{X})$. Hence $\bar{U} \cap \bar{V} = \emptyset$. Since both $\bar{U}$ and $\bar{V}$ are invariant under $G(\tilde{X})$, $\pi(\bar{U})$ and $\pi(\bar{V})$ are disjoint open sets containing $\pi(\tilde{x})$ and $\pi(\tilde{y})$, respectively. Thus $\tilde{X}/G(\tilde{X})$ is Hausdorff.

Now we define the complex structure. Pick $\tilde{x} \in \tilde{X}$ and consider $\pi(\tilde{x}) \in \tilde{X}/G(\tilde{X})$. Recall in the proof of Proposition 2.26 we showed there exists some neighborhood U of $\tilde{x}$ such that $g(U) \cap U = \emptyset$ for all non-identity elements $g \in G(\tilde{X})$. Then this implies $\pi|_U$ is injective, and $\pi|_U$ is by definition surjective onto its image. As we showed in the proof of Theorem 2.22, π is also continuous and open, and hence a homeomorphism.

Now we can always intersect U with another set and preserve the property action by $G(\tilde{X})$ is disjoint, so WLOG assume $\varphi : U \rightarrow \mathbb{C}$ is a chart for $\tilde{X}$. Then define a chart $\phi : \pi(U) \subseteq \tilde{X}/G(\tilde{X}) \rightarrow \mathbb{C}$ on the quotient by $\phi = \varphi \circ (\pi|_U)^{-1}$. Similar to Lemma 2.29 we can check the transition maps are compatible so the collection of all such charts defines a complex atlas.

To show π is locally conformal, note that π is a local homeomorphism and by taking charts on both sides we have $\phi_\alpha \circ \pi \circ \varphi_\beta^{-1} = \varphi_\alpha \circ (\pi|_{U_\alpha})^{-1} \circ \pi \circ \varphi_\beta^{-1} = \varphi_\alpha \circ \varphi_\beta^{-1}$ which is holomorphic on the domain where it is defined. Since π is locally bijective and holomorphic, it is locally conformal.

Uniqueness also follows by the same argument as in Lemma 2.29. $\square$

Theorem 2.35. *Let X be Riemann surface and $\tilde{X}$ its universal cover. $\tilde{X}/G(\tilde{X})$ is conformally equivalent to X .*

Proof. As before we have that $\tilde{p} : \tilde{X}/G(\tilde{X}) \rightarrow X$ is a homeomorphism, where $\tilde{p} = p \circ \pi$. But since both π and p are locally conformal, by choosing a neighborhood U such that $p|_U$ and $\pi|_U$ are conformal, we can write $\tilde{p}|_{\pi(U)} = p|_U \circ (\pi|_U)^{-1}$, so $\tilde{p}$ is locally conformal. Since it is also a homeomorphism and thus bijective, $\tilde{p}$ is conformal. $\square$

To summarize we have shown that:

- Every Riemann surface has a universal cover, through which we can classify them as elliptic, parabolic, or hyperbolic.
- The group of deck transformations acts properly discontinuously on the universal cover.
- Every Riemann surface is conformally equivalent to the quotient of its universal cover by its group of deck transformations.

3. FUCHSIAN MODELS OF HYPERBOLIC RIEMANN SURFACES

3.1. Fuchsian Groups and Fuchsian Models. We now look at the hyperbolic case more closely. The reason we call Riemann surfaces hyperbolic when their universal cover is the upper half plane is because the upper half plane is a model for the hyperbolic plane.

Recall a few basic facts about Möbius transformations:

Proposition 3.1.

(i) *The group of Möbius transformations is given by*

$$\text{Möb} = \left\{ z \mapsto \frac{az + b}{cz + d} : a, b, c, d \in \mathbb{C}, ad - bc \neq 0 \right\}$$

(ii) *The automorphisms of the Riemann sphere $\hat{\mathbb{C}}$ are exactly the Möbius transformations.*

(iii) *Möbius transformations preserve Riemann sphere circles.*

Now recall the following facts about the Poincaré disk model:

Proposition 3.2.

(i) *The group of conformal automorphisms of $\mathbb{D}$ is given by*

$$\text{Aut}(\mathbb{D}) = \left\{ z \mapsto e^{i\theta} \frac{z - a}{1 - \bar{a}z} : \theta \in \mathbb{R}, a \in \mathbb{D} \right\} \subseteq \text{Möb}$$

(ii) $d(z, w) = 2 \tanh^{-1} \left| \frac{z - w}{1 - \bar{z}w} \right|$

defines the hyperbolic metric in the disk, which is invariant under the automorphisms of the disk.

(iii) *The hyperbolic circles are exactly the Euclidean circles in the disk.*

Using a Riemann map from $\mathbb{H}$ to $\mathbb{D}$

$$\phi(z) = \frac{z - i}{z + i}$$

we can then map everything to the upper half plane model:

Proposition 3.3.

(i) *The group of conformal automorphisms of $\mathbb{H}$ is given by*

$$\text{Aut}(\mathbb{H}) = \left\{ z \mapsto \frac{az + b}{cz + d} : a, b, c, d \in \mathbb{R}, ad - bc = 1 \right\}$$

Hence by putting the coefficients into a matrix, the group can be identified with $\text{PSL}(2, \mathbb{R})$.

(ii) $d(z, w) = 2 \tanh^{-1} \left| \frac{z - w}{z - \bar{w}} \right|$

defines the hyperbolic metric in the upper half plane, and invariant under automorphisms.

(iii) *The hyperbolic circles are exactly the Euclidean circles in the upper half plane.*

Corollary 3.4. *The upper half plane with the hyperbolic metric induces the same topology as the Euclidean metric.*

Since these are just calculations, we omit the proof.

Note that $\mathrm{PSL}(2, \mathbb{R})$ is a Lie group, which induces a topology on it. This can be seen by using the implicit function theorem to show $\mathrm{SL}(2, \mathbb{R})$ is a Lie group, and then quotienting out by $\{\pm I\}$, which is a closed subgroup.

Recall that the deck transformations on a Riemann surface are a subgroup of the conformal automorphisms which act properly discontinuously (and we mentioned it was possible to show they were discrete). This motivates the definition of Fuchsian groups:

Definition 3.5. A **Fuchsian group** is a discrete subgroup of $\mathrm{PSL}(2, \mathbb{R})$.

We now characterize them:

Theorem 3.6. *Let Γ be a subgroup of $\mathrm{PSL}(2, \mathbb{R})$. Then Γ is a Fuchsian group iff Γ acts properly discontinuously on $\mathbb{H}$.*

We will need to establish some lemmas first.

Lemma 3.7. *Let $z_0 \in \mathbb{H}$ and $K \subseteq \mathbb{H}$ be compact. Then the set $E = \{T \in \mathrm{PSL}(2, \mathbb{R}) : T(z_0) \in K\}$ is compact.*

Proof. By identifying $\mathrm{PSL}(2, \mathbb{R})$ with $\mathrm{Aut}(\mathbb{H})$, we can define $\psi : \mathrm{SL}(2, \mathbb{R}) \rightarrow \mathrm{PSL}(2, \mathbb{R})$ by

$$\psi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = T, \quad T(z) = \frac{az + b}{cz + d}$$

Since taking the negative coefficients gives the same map, this is effectively quotienting out by $\{\pm I\}$. This map is continuous by definition and if we show

$$E_1 = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{R}) : \frac{az_0 + b}{cz_0 + d} \in K \right\}$$

is compact, then $E = \psi(E_1)$ is compact. We show E_1 is compact, by embedding $\mathrm{SL}(2, \mathbb{R})$ in $\mathbb{R}^4$ and showing E_1 is closed and bounded.

Note that since $\mathrm{PSL}(2, \mathbb{R})$ acts on $\mathbb{H}$ by conformal maps, we get a continuous map $\beta : \mathrm{SL}(2, \mathbb{R}) \rightarrow \mathbb{H}$ defined by $\beta(A) = \psi(A)(z_0)$. It follows that $E_1 = \beta^{-1}(K)$, so E_1 is closed.

To show E_1 is bounded, we show the coefficients are bounded. Since K is bounded there exists $M_1 > 0$ such that

$$\left| \frac{az_0 + b}{cz_0 + d} \right| \leq M_1$$

for all corresponding matrices in E_1 . But since K and $\mathbb{H}^C$ are both compact, they have positive distance, so there exists $M_2 > 0$ such that

$$\mathrm{Im} \left(\frac{az_0 + b}{cz_0 + d} \right) \geq M_2$$

But using the fact that $ad - bc = 1$ (since the matrix is in $\mathrm{PSL}(2, \mathbb{R})$) we get

$$\frac{az_0 + b}{cz_0 + d} = \frac{ac|z_0|^2 + adz_0 + bc\bar{z}_0 + bd}{|cz_0 + d|^2} \implies \mathrm{Im} \left(\frac{az_0 + b}{cz_0 + d} \right) = \frac{\mathrm{Im}(z_0)}{|cz_0 + d|^2}$$

This gives the bounds

$$|cz_0 + d| \leq \sqrt{\frac{\mathrm{Im}(z_0)}{M_2}} \implies |az_0 + b| \leq M_1 \sqrt{\frac{\mathrm{Im}(z_0)}{M_2}}$$

which then implies all the coefficients are bounded, and hence E_1 is compact, so E is compact. $\square$

Lemma 3.8. *Let Γ be a subgroup of $\mathrm{PSL}(2, \mathbb{R})$ acting properly discontinuously on $\mathbb{H}$, and $p \in \mathbb{H}$ be fixed by some element of Γ . Then there is a neighborhood W of p for which no other point of W is fixed by an element of Γ other than the identity.*

Proof. Suppose $T(p) = p$ for some non-identity $T \in \Gamma$ and suppose for contradiction in every neighborhood of p there are fixed points of Γ , so there exists a sequence $p_n \rightarrow p$ and a sequence of transformations $T_n \in \Gamma$ such that $T_n(p_n) = p_n$. Now consider a closed hyperbolic disk $\overline{B(p, 3\varepsilon)}$. Since this is compact in the Euclidean topology, it is also compact in the topology induced by the hyperbolic metric. We claim the set

$$\{T \in \Gamma : T(p) \in \overline{B(p, 3\varepsilon)}\}$$

is finite. Suppose there were infinitely many distinct such S_i in the set. By compactness, passing to a subsequence if necessary, let $S_i(p) \rightarrow q$ for some $q \in \mathbb{H}$. Since Γ acts properly discontinuously, q has a neighborhood U for which $\{T \in \Gamma : T(U) \cap U \neq \emptyset\}$ is finite. For all sufficiently large i we have $S_i(p) \in U$. Fix some j sufficiently large, and also let k be sufficiently large. Let $R_k := S_j \circ S_k^{-1}$. Then note $R_k(S_k(p)) = S_j(p)$. This means $S_j(p) \in R_k(U) \cap U$, so $R_k(U) \cap U \neq \emptyset$. But we can do this for infinitely many k , and since all the S_k 's are distinct, so are the R_k 's, a contradiction.

If T_n were eventually constant, i.e. $T_n = \tilde{T}$ for some $\tilde{T} \in \Gamma$ and sufficiently large n , then since $p_n \rightarrow p$, $\tilde{T}$ would have an accumulation point, and by the identity theorem must be the identity. Thus assume T_n takes on infinitely many elements of Γ . Then by the finiteness of the above set we must have $d(T_n(p), p) > 3\varepsilon$ and $d(p_n, p) < \varepsilon$ for sufficiently large n . Then since the hyperbolic metric is invariant under Γ , we have

$$d(T_n(p), p) \leq d(T_n(p), T_n(p_n)) + d(T_n(p_n), p) = d(p, p_n) + d(p_n, p) < 2\varepsilon$$

a contradiction. □

Proof of Theorem 3.6. ($\implies$) Let $z \in \mathbb{H}$ and $K \subseteq \mathbb{H}$ be compact. Then $\{T \in \Gamma : T(z) \in K\} = \{T \in \mathrm{PSL}(2, \mathbb{R}) : T(z) \in K\} \cap \Gamma$. The first set is compact by Lemma 3.7 and the second set is discrete by assumption. Hence the set is finite, and Γ acts properly discontinuously.

($\impliedby$) Suppose Γ acts properly discontinuously but is not discrete. By Lemma 3.8, choose some $s \in \mathbb{H}$ which is not fixed by any element in Γ other than the identity. Since Γ is not discrete, there exists a sequence $\{T_n\}_{n \in \mathbb{N}} \subseteq \Gamma$ of distinct elements such that $T_n \rightarrow \mathrm{id}$. (Not discrete means there is an accumulation point, but we can always use the group structure to move it back to the identity.) Then $T_n(s) \rightarrow s$, but since s is only fixed by the identity, $T_n(s)$ must all be different from s . But this means any neighborhood U of s contains points of its orbit, and so there are infinitely many distinct T_n 's for which $T_n(U) \cap U \neq \emptyset$, and Γ does not act properly discontinuously, a contradiction. □

Corollary 3.9 (Fuchsian Models for Hyperbolic Riemann Surfaces). *Let X be a hyperbolic Riemann surface. Then X is conformally equivalent to $\mathbb{H}/\Gamma$ where Γ is some Fuchsian group.*

3.2. Further Discussion. At this point there are two main questions that have not been addressed:

- (1) Why is the hyperbolic case more interesting than the elliptic and parabolic ones?
- (2) How does the geometry of the hyperbolic plane come into the Riemann surface?

We discuss this informally and briefly.

For the first question, recall that the group of deck transformations acts freely, so there cannot be any fixed points. When the universal cover is $\hat{\mathbb{C}}$, the group of conformal automorphisms are exactly the Möbius transformations. But it is well known that Möbius transformations have at least one fixed point, so there the deck transformations must be trivial, and the only elliptic Riemann surface is $\hat{\mathbb{C}}$ itself.

It is also a standard fact in complex analysis that the only conformal automorphisms of $\mathbb{C}$ are of the form $z \mapsto az + b$. Requiring no fixed points forces $a = 1$, the deck transformations end up being a discrete subgroup of the group of translations. It can then be shown the only possible such subgroups are trivial, cyclic, or a rank 2 lattice, respectively giving rise to $\mathbb{C}$, a cylinder, and a torus.

On the other hand, Fuchsian groups are much richer in possibilities. I have not read too much on this topic, but I believe it has to do with the structure of the Lie algebra of $\mathrm{PSL}(2, \mathbb{R})$.

For the second question, recall the example of a torus, where studying the torus can be reduced to studying a fundamental parallelogram. Similarly, hyperbolic Riemann surfaces also have fundamental domains, and these are most naturally described with hyperbolic geodesics (see Katok). This accomplishes our original goal in the introduction of wanting to reduce studying Riemann surfaces to studying some more basic region.

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REFERENCES

- [1] S. Katok, *Fuchsian Groups*, University of Chicago Press, Chicago, 1992.
- [2] O. Forster, *Lectures on Riemann Surfaces*, Graduate Texts in Mathematics, vol. 81, Springer-Verlag, New York, 1981.
- [3] T. Tao, *246C notes 1: Meromorphic functions on Riemann surfaces, and the Riemann–Roch theorem*, 2018. Available at <https://terrytao.wordpress.com/2018/03/28/246c-notes-1-meromorphic-functions-on-riemann-surfaces-and-the-riemann-roch-theorem/>.
- [4] M. Bonk, *Complex Analysis Lecture Notes*, unpublished.
- [5] A. Hatcher, *Algebraic Topology*, Cambridge University Press, Cambridge, 2002.
- [6] B. Petri, *Teichmüller Theory*, lecture notes, 2019. Available at https://webusers.imj-prg.fr/~bram.petri/t_tt/Teich190704.pdf.
- [7] B. Simon, *Representations of Finite and Compact Groups*, Graduate Studies in Mathematics, vol. 10, American Mathematical Society, Providence, RI, 1996.